Summary in Graph

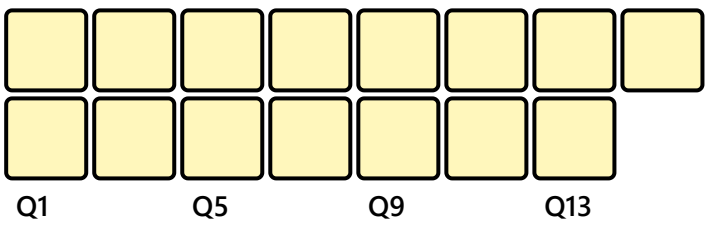
Exam Summary (GO Classes CS | CO and Architecture | Cache Memory | Weekly Quiz 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (22) out of 222 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 CO & Architecture

Assume a system with:

- 8-bit memory addresses
- direct-mapped cache

- 2-byte cache block size
- 4 cache lines

How many bytes of data can be stored in the cache? Do not include metadata like the tag or valid bit.

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

Assume a system with:

- 8-bit memory addresses
- direct-mapped cache
- 2-byte cache block size
- 4 cache lines

Determine the number of tag bits.

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | CO & Architecture

In a direct-mapped cache, how are the different bits of the memory address used? (Select all correct options.)

A. The tag bits are used to uniquely identify which block of main memory is currently stored in a cache line.

B. The index bits are used to select which cache line a memory block maps to.

C. The byte offset bits are used to select the specific byte within a cache block.

D. The tag bits are used to select which cache line a memory block maps to.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

Assume a system with:

- 8-bit memory addresses
- 2-byte cache block size
- The cache is 2-way set associative with a total of 4 sets

Determine the number of tag bits.

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

LRU is an effective cache replacement strategy, primarily because programs

A. exhibit locality of reference

B. usually have small working sets

C. read data much more frequently than write data

D. always access memory addresses in strictly increasing order

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #6 Multiple Select Type Award: 2 Penalty: 0 CO & Architecture

Assume a system with:

- 8-bit memory addresses
- direct-mapped cache
- 2-byte cache block size
- 4 cache lines

Which of the following pairs of memory addresses belong to different memory blocks but map to the same cache line? (Addresses are specified in base-2.)

A. 00000000 and 00001000

B. 00100101 and 01100101

C. 01010010 and 01010011

D. 01101010 and 01101100

Your Answer: Correct Answer: A;B Not Attempted Time taken: 00min 00sec Discuss

Q #7 Numerical Type Award: 2 Penalty: 0 CO & Architecture

Assume a system with:

- 8-bit memory addresses
- direct-mapped cache
- 2-byte cache block size
- 4 cache lines

The following memory operations (R: read, W : write) are performed. Addresses are specified in base-2.

1. R00101010

2. W00101011

3. R11001100

4. R11001010

5. R11001011

6. W00111101

7. R00111100

8. W00101100

Each operation above is either a hit or a miss, which indicates whether the data is found in the cache or not. All cache lines are initially empty.

What is the total number of cache misses that occur during this sequence?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume a system with:

- 8-bit memory addresses
- 2-byte cache block size
- The cache is 2-way set associative with a total of 4 sets

The following memory operations (R: read, W: write) are performed. Addresses are specified in base-2.

1. R00011010

2. W00011011

3. R11111000

4. R11111010

5. R01101000

6. W00001001

7. R00000000

8. W00011010

Each operation above is either a hit or a miss, which indicates whether the data is found in the cache or not. All cache lines are initially empty. When a block is evicted, select the victim using the LRU (Least Recently Used) replacement policy.

What is the total number of cache misses that occur during this sequence?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

If increasing the block size of a cache improves performance, it is primarily because programs

A. exhibit temporal locality

B. exhibit spatial locality

C. usually have small working sets

D. read data much more frequently than write data

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Which of the following statements is/are FALSE?

I: In set associative mapping, if set size is reduced to 1, it reduces to fully associative mapping.

II: In set associative mapping, if only one set is present, it reduces to direct mapping.

A. I Only

B. II Only

C. Both I and II

D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a system with a block size of p words. What is the range of words present in the kth block of main memory?

A. $k \cdot p \text{ to } (k+1) \cdot p - 1$

B. $k \cdot p^2 \text{ to } (k+1) \cdot p^2$

C. $k \cdot 2^p \text{ to } (k+1) \cdot 2^p - 1$

D. $k \cdot 2^p \text{ to } k \cdot 2^{p+1} - 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer has a 512 KB , 8 -way set associative data cache with a block size of 32 B. The 1 point processor sends 32-bit addresses to the cache controller. Each cache tag directory entry contains, in addition to the address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

what will be the number of tag bits?

A. 11

B. 16

C. 21

D. 12

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer's processor sends 32 bit addresses to the cache controller. It has a $512\text{KByte}, 8$-way set associative, write-back data cache with a block size of 32 Bytes. In addition to the address tag, each cache tag directory entry contains 3 valid bits and 1 modified bit. Find the size of the cache tag directory.

A. 212 Kbits

B. 320 Kbits

C. 160 Kbits

D. 120 Kbits

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is:

$$8, 12, 0, 12, 8$$

A. 2

B. 3

C. 4

D. 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is ___

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

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